of sex determination, the alphabets 'Z' and 'W' are used here instead of X and Y respectively. Heterogametic females are of two types, ZO-ZZ type (eg. Moths, butterflies and domestic chickens) and ZW-ZZ type (eg. Gypsy moth, fishes, reptiles and birds).

## Sex determination in human beings

Genes determining sex in human beings are located on two sex chromosomes, called allosomes. In mammals, sex determination is associated with chromosomal differences between the two sexes, typically XX females and XY males. 23 pairs of human chromosomes include 22 pairs of autosomes (44A) and one pair of sex chromosomes (XX or XY). Females are homogametic producing only one type of gamete (egg), each containing one X chromosome while the males are heterogametic producing two types of sperms with X and Y chromosomes. An independently evolved XX: XY system of sex chromosomes also exist in *Drosophila* **(Fig. 4.2)**.

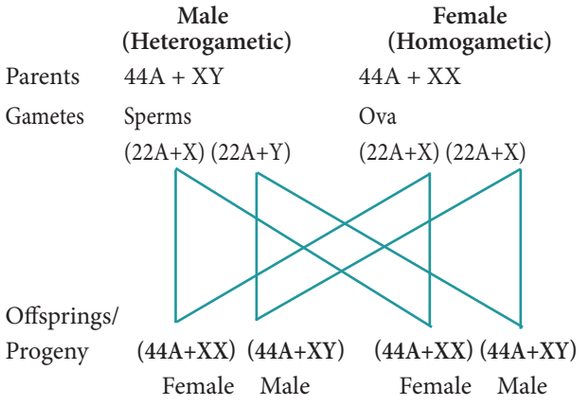



**Fig. 4.2 Sex determination in human beings**

## The Y Chromosome and Male Development

Current analysis of Y chromosomes has revealed numerous genes and regions with potential genetic function; some genes with or without homologous counterparts are seen on the X. Present at both ends of the Y chromosome are the pseudoautosomal regions (PARs) that are similar with regions on the X chromosome which synapse and recombine during meiosis. The remaining 95% of the Y chromosome is referred as the Non - combining Region of the Y (NRY). The NRY is divided equally into functional genes (euchromatic) and non functional genes (heterochromatic). Within the euchromatin regions, is a gene called **Sex determining region Y (SRY)**. In humans, absence of Y chromosome inevitably leads to female development and this SRY gene is absent in X chromosome. The gene product of SRY is the testes determining factor (TDF) present in the adult male testis.

- X-Chromosome was discovered by Henking (1891)
- Y-Chromosome was discovered by Stevens (1902)

### 4.4.1 Dosage compensation - Barr body

In 1949, Barr and Bertram first observed a condensed body in the nerve cells of female cat which was absent in the male. This condensed body was called sex chromatin by them and was later referred as **Barr body**. In the XY chromosomal system of sex determination, males have only one X chromosome, whereas females have two. A question arises: how does the organism compensate for this dosage differences between the sexes? In mammals the necessary dosage compensation is accomplished by the inactivation of one of the X chromosome in females so that both males and females have only one functional X chromosome per cell.

Mary Lyon suggested that Barr bodies represented an inactive chromosome, which in females becomes tightly coiled into a heterochromatin, a condensed and visible form of chromatin (Lyon's hypothesis). The number of Barr bodies observed in cell was one less than the number of X-Chromosome. XO females have no Barr body, whereas XXY males have one Barr body.



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